THE UNIVERSITY OF LAHORE

DEPARTMENT OF COMPUTER SCIENCE | MACHINE LEARNING

PREDICTIVE MODELING OF EMPLOYEE ATTRITION

Updated study using a 10,000-record public HR Analytics benchmark

10,000	98.25%	0.9802
Records	Test accuracy	Test ROC-AUC

PROJECT STUDENTS

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Research update / 11 November 2026

Educational benchmark only - not a production HR decision system

RESEARCH BRIEF

Executive summary

A refreshed model comparison based on the supplied Colab outputs and exported results workbook.

10,000
Employee records
9 model inputs

98.25%
Test accuracy
Random Forest wins

90.36%
Leaver recall
32 of 332 missed

Primary findings

Random Forest achieved the strongest validation F1-score (94.74%) and was selected before opening the final 20% test set. On 2,000 held-out rows, it correctly classified 1,965 observations: 1,665 non-leavers and 300 leavers.

The model made 35 errors overall, including 32 false negatives (employees who left but were predicted to stay). Precision for the leaving class reached 99.01%, while recall reached 90.36%.

What this update changes

Important comparison boundary

The earlier IBM experiment used 1,470 rows and 35 columns. This update uses a different public HR Analytics dataset with 10 columns. Any accuracy difference cannot be attributed solely to a larger sample size.

REPORT SCOPE

Data provenance, experimental split, three-classifier validation, final-test diagnostics, interpretation, limitations, and reproducibility. Statistics are from the attached experiment results; no new model run was used to replace them.

Dataset and provenance

Source: publicly shared HR Analytics data; output file prepared for the 10,000-row project.

DATA PROPERTY	VALUE
Original public benchmark	14,999 rows
Deduplicated predictor combinations	11,991 unique rows
Selected sample	10,000 rows
Input features	9 (7 numeric + 2 categorical)
Target variable	Attrition: Yes / No
Observed attrition rate	16.6% (1,660 of 10,000)
Missing values	0
Exact duplicate input combinations	0



Provenance limitation

This is an educational/public benchmark. Its source does not establish verified real employee identities or cross-company representativeness.

Training and evaluation protocol

Stratified three-way split, all preprocessing fitted only on the training subset.



End-to-end workflow

01

Validation-safe split

80% development / 20% untouched test; development split again for 64% train / 16% validation.

02

Preprocessing pipeline

Numeric median imputation + StandardScaler; categorical most-frequent imputation + one-hot encoding.

03

Model comparison

Logistic Regression, Random Forest, SVM trained on the same training partition. F1 on validation determines winner.

04

Final holdout

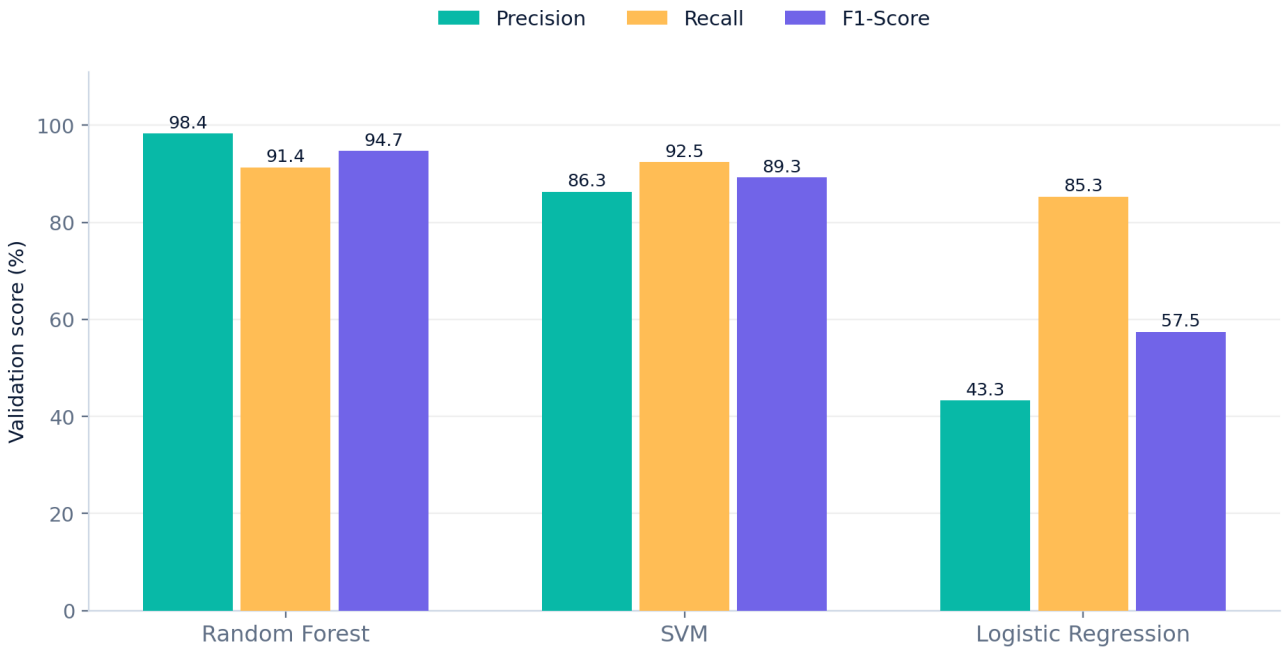
Best classifier refitted on full 80% development partition. Final 2,000 records evaluated once.

Controls against leakage

Scaling and category encoding are inside scikit-learn pipelines. No synthetic oversampling (SMOTE) is applied in this updated run.

Which classifier performed best?

Metrics use Attrition = Yes as the positive class; validation sample size: n = 1,600.



MODEL	ACC	PREC	RECALL	F1	AUC
Random Forest	98.31%	98.38%	91.35%	94.74%	0.9860
SVM	96.31%	86.32%	92.48%	89.29%	0.9782
Logistic Regression	79.00%	43.32%	85.34%	57.47%	0.8486

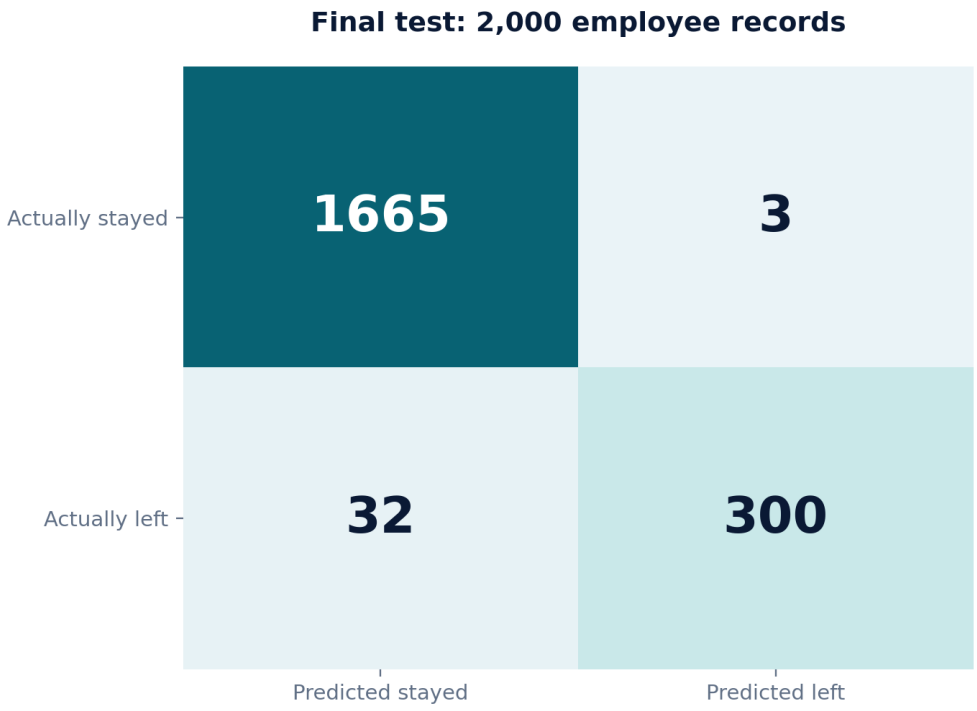
Selection rule: highest validation F1

Random Forest leads on F1 (94.74%) and precision (98.38%). SVM has slightly higher recall (92.48% vs 91.35%), but lower precision and F1.

Performance on unseen holdout

Only the winning model - Random Forest - was evaluated on the 2,000-row final test set.

98.25%	99.01%	90.36%	94.49%
Accuracy	Precision	Recall	F1-score



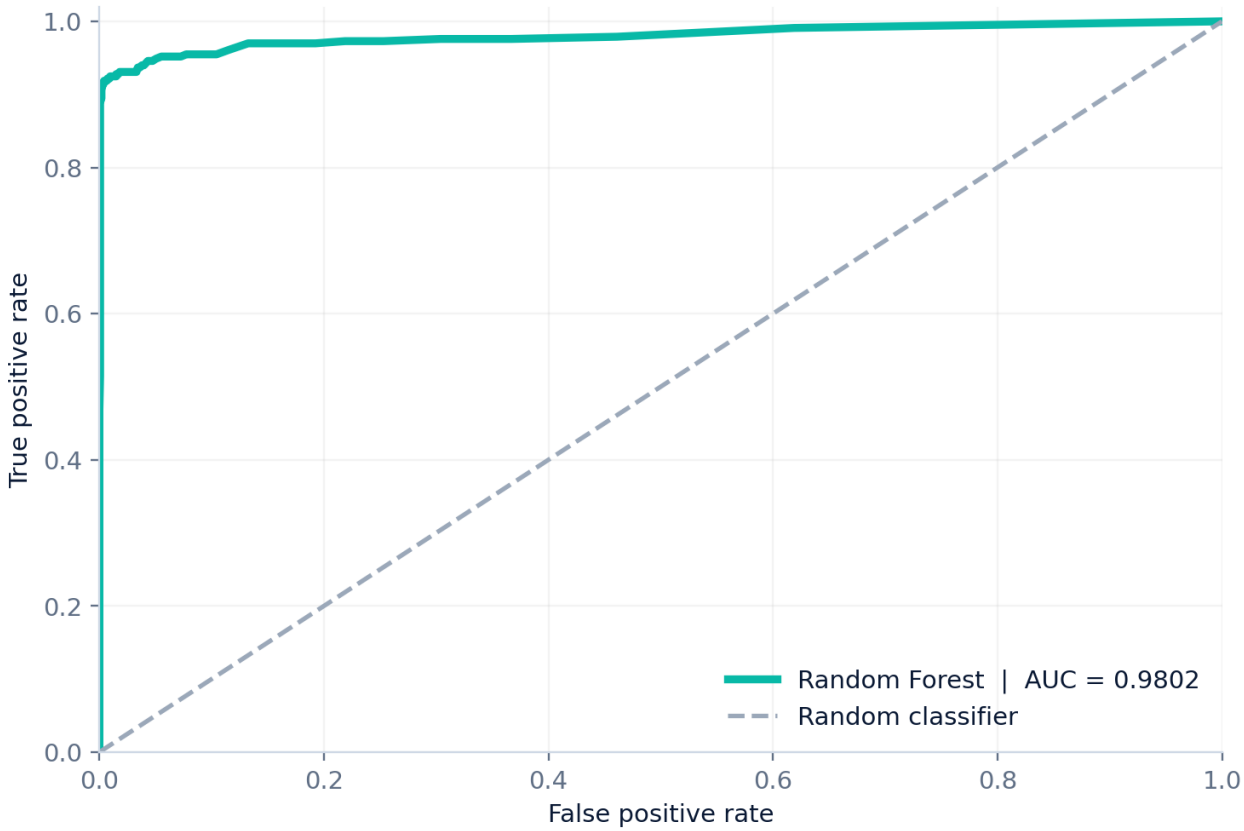
TRUE NEGATIVES	FALSE POSITIVES	FALSE NEGATIVES	TRUE POSITIVES
1,665	3	32	300

Why recall matters

32 actual leavers were classified as staying. For an attrition-screening use case, these misses should be weighed alongside the high 98.25% overall accuracy.

ROC-AUC and classification quality

The score reflects ranking performance at multiple thresholds, not real-world calibration.



0.9802

Test ROC-AUC

Strong in-source ranking

99.82%

Specificity

1,665 / 1,668

95.09%

Balanced accuracy

Mean of class recalls

Interpretation

The model separates leaving versus staying cases well within this benchmark. ROC-AUC does not establish that risk probabilities are calibrated, transferable to another organization, or safe for personnel decisions.

Original IBM study vs updated run

These are two different experiments and do not form a controlled dataset-size ablation.

FACTOR	ORIGINAL STUDY	UPDATED EXPERIMENT
Dataset	IBM HR Analytics	Public HR Analytics benchmark
Records	1,470	10,000
Dataset columns	35	10
Models	LR, SVM, RF	LR, SVM, RF
Best model	Tuned SVM	Random Forest
Reported test accuracy	86.73%	98.25%
Reported leaver recall	34%	90.36%
Reported AUC	0.8108	0.9802

Critical scientific interpretation

The updated run uses different features, a different label distribution, and a different benchmark. Its higher score is not causal evidence that using 10,000 instead of 1,470 observations improves accuracy by 11.52 percentage points.

WHY THE UPDATE IS STILL USEFUL

It demonstrates a reproducible three-model workflow, a larger evaluation sample, separate model selection and test phases, and detailed minority-class metrics that improve over reporting accuracy alone.

Limitations and practical safeguards

These constraints are essential in an academic report about HR-related predictive systems.

Dataset authenticity

Public benchmark should not be presented as validated real organizational personnel data.

External validity

Random holdout from the same source cannot estimate performance across employers or later years.

Feature coverage

Only nine model inputs. No age, gender, or protected-class evaluation; no fairness claims are possible.

Class distribution

Only 16.6% of records are leavers; accuracy can hide poor detection of the minority class.

False negatives

32 missed leavers in the final test show that 98.25% accuracy is not 98.25% sensitivity.

Decision governance

Never use predictions alone for firing, promotion or adverse employment decisions.

Reproducibility

Version differences and source data sampling can change results; disclose processing choices.

Research framing

Report these figures as benchmark results, not a production-ready retention prediction service.

Files, code, and source references

The associated Colab notebook and source package reproduce the described analysis.

Artifacts

Google Colab notebook

Employee_Attrition_10000_Google_Colab.ipynb

Input workbook

Employee_Attrition_10000_Training_Dataset.xlsx

Results workbook

Employee_Attrition_Results.xlsx

Code package

Employee_Attrition_GitHub_Ready.zip

Reproduction checklist

* Open the Colab notebook and upload the 10,000-row Excel workbook.
* Run all cells in order with random_state = 42.
* Confirm 6,400 / 1,600 / 2,000 training, validation, and test samples.
* Choose the best model by validation F1.
* Evaluate the selected model once on the untouched test set.

SOURCE LINKS

Public dataset reference: https://github.com/RuchitaGarde/Human-Resources-Analytics

Implementation library: https://scikit-learn.org/stable/

This report is derived from the supplied workbook, screenshots, notebook, and original IBM-based project article.